

A Counterexample for Rectangular Products and a note on the asymmetric-channel literature

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Abstract

We consider the direct rectangular analogue of the central-residue construction for k -separated families in products of chains. We give an elementary counterexample for every $k \geq 2$, already when $n = 2k + 1$, the first dimension not covered by the two-block argument for $k \geq n/2$. We also record how the problem fits into Kovačević's framework for asymmetric-error-detecting codes and note that this reference is not currently present in the bibliography of the main paper.

1 The natural rectangular construction

Let

$$P(\mathbf{b}) = \prod_{i=1}^n \{0, 1, \dots, b_i\}, \quad \rho(x) = \sum_{i=1}^n x_i, \quad T = \sum_{i=1}^n b_i.$$

A family $\mathcal{F} \subseteq P(\mathbf{b})$ is *k-separated* if, whenever $x < y$ coordinatewise are distinct members of $\mathcal{F}$, they differ in more than k coordinates.

Write $b_{(1)} \geq \dots \geq b_{(n)}$ for the decreasing rearrangement of the bounds, and put

$$D_k = \sum_{i=1}^k b_{(i)}, \quad M = D_k + 1.$$

The direct analogue of the two residue families in the equal-sided cube is

$$\mathcal{A}^- = \{x : \rho(x) \equiv \lfloor T/2 \rfloor \pmod{M}\}, \quad \mathcal{A}^+ = \{x : \rho(x) \equiv \lceil T/2 \rceil \pmod{M}\}.$$

Both are k -separated. Indeed, if $x < y$ differ in at most k coordinates, then

$$1 \leq \rho(y) - \rho(x) \leq D_k < M.$$

Coordinatewise reflection $x_i \mapsto b_i - x_i$ exchanges $\mathcal{A}^-$ and $\mathcal{A}^+$, so they have the same cardinality.

When $k \geq n/2$, one has $D_k \geq T/2$. Thus $M > T/2$, and these residue families reduce to the lower and upper central layers. This is precisely the range in which the two-block symmetric-chain argument extends to rectangular products. The construction below shows that the analogous statement fails immediately outside that range.

2 An infinite family of counterexamples

For $r \in \mathbb{Z}$, write

$$L_r = \{x \in P(\mathbf{b}) : \rho(x) = r\}.$$

Theorem 1. *For every $k \geq 2$, there is a rectangular product in dimension $2k + 1$ for which a k -separated family is strictly larger than both central-residue families $\mathcal{A}^-$ and $\mathcal{A}^+$.*

Proof. Take $k + 1$ coordinates of bound $k + 1$ and k coordinates of bound $k + 2$:

$$P = \{0, \dots, k + 1\}^{k+1} \times \{0, \dots, k + 2\}^k.$$

The sum of the k largest bounds and the total rank are

$$D := D_k = k(k + 2), \quad T = (k + 1)^2 + k(k + 2) = 2D + 1.$$

Consequently $M = D + 1$, and the lower central-residue family is exactly

$$\mathcal{A}^- = L_D \cup \{\mathbf{b}\},$$

where $\mathbf{b}$ is the top vertex of P .

For each of the k coordinates of bound $k + 2$, let $u_j = \mathbf{b} - e_j$ be the corresponding coatom, and define

$$\mathcal{B} = L_D \cup \{u_1, \dots, u_k\}.$$

The layer L_D is an antichain, and the u_j are pairwise incomparable. It remains only to rule out a forbidden comparison between the two parts.

Suppose that $x \in L_D$ satisfies $x \leq u_j$ and differs from u_j in at most k coordinates. Since $\rho(u_j) = 2D$, the total decrease from u_j to x is D . On the other hand, the sum of the k largest coordinates of u_j is

$$(k - 1)(k + 2) + (k + 1) = D - 1.$$

No change in at most k coordinates can therefore decrease its rank by D , a contradiction. Hence $\mathcal{B}$ is k -separated. Finally,

$$|\mathcal{B}| = |L_D| + k > |L_D| + 1 = |\mathcal{A}^-| = |\mathcal{A}^+|.$$

□

Remark 2 (First instance). For $k = 2$, take bounds $(3, 3, 3, 4, 4)$. Here $D = 8$, $M = 9$, and $T = 17$. The coefficient of z^8 in

$$(1 + z + z^2 + z^3)^3(1 + z + z^2 + z^3 + z^4)^2$$

is

$$\binom{12}{4} - 3\binom{8}{4} - 2\binom{7}{4} + 3\binom{4}{4} = 218.$$

Thus each central-residue family has size 219, whereas

$$L_8 \cup \{(3, 3, 3, 3, 4), (3, 3, 3, 4, 3)\}$$

is 2-separated and has size 220. A maximum-independent-set computation also gives optimum 220, but that upper bound is not needed for the counterexample.

Remark 3 (Uniqueness). Even in the range $k \geq n/2$, the rectangular cardinality result does not generally have the uniqueness conclusion of the equal-sided theorem. For $n = k = 2$ and bounds $(1, 2)$, the two central layers and

$$\{(1, 0), (0, 2)\}$$

are three distinct maximum antichains of size 2.

3 Relation with asymmetric-error-detecting codes

Kovačević [1] studies codes in $\{0, \dots, q - 1\}^n$ that detect coordinatewise-increasing errors subject to three possible bounds: per-coordinate amplitude a , Hamming support h , and total increase t . The equal-sided problem in the main paper is the finite, non-cyclic specialization

$$q = d + 1, \quad a = d, \quad h = k, \quad t = dk.$$

Here the amplitude and total-increase bounds are redundant, so only the support bound remains. Thus Kovačević’s *model* contains the problem, but his finite-alphabet optimality theorems do not contain the main paper’s result for $k > 1$. They establish the modular construction on the infinite lattice and in several different finite parameter regimes, while the unrestricted finite problem is left open. The weighted-chain theorem for $d = 1, 2$ therefore supplies an exact finite-boundary result not furnished by that paper.

At the time of writing this note, Kovačević [1] is not cited in the bibliography of the main paper. It would be natural to add it to the coding-theory discussion, while making clear that it provides the ambient framework rather than the theorem proved here.

References

- [1] M. Kovačević, On error detection in asymmetric channels, *IEEE Communications Letters* 21 (2017), no. 9, 1933–1936. <https://arxiv.org/abs/1706.04540>.